New Product Development – A case study at Dow Corning

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Agenda

- *About Dow Corning*
- *History of Product Development*
- *Case Study: Deck Protector*
- *Lessons learned*

History of Dow Corning

Dow Corning was formally established in 1943 as a joint venture between the American conglomerates Dow Chemical and Corning Glass to explore the potential of silicone and was a manufacturer of products for use by the U.S. military in World War II. The company began operating its first plant, in Midland, MI, in 1945.

Following the December 11, 2015 announcement that it would merge with DuPont, Dow also announced on the same day that it had reached a deal to acquire Corning's stake in Dow Corning in exchange for $4.8 billion in cash and Corning gaining a roughly 40% stake in Hemlock Semiconductor Corporation. Dow Chemical assumed full ownership of Dow Corning on June 1, 2016. The company changed its name to Dow Silicones Corporation in 2018.

Products

Dow Silicones markets over 7000 products, including various sealants, adhesives, rubbers, lubricants, silicon oils and solvents. Around 2,100 of these are available through the Dow Corning online-only distributor Xiameter, including fluids, gels, resins. The range of industries targeted by Dow Corning products spans from electronics and automotive to construction, healthcare and others. In recent years, the company has expanded production of solar cells, particularly through its majority stake in Hemlock Semiconductor Corporation, which accounts for a polysilicon franchise worth over $1 billion.

Dow Corning’s Transformation program

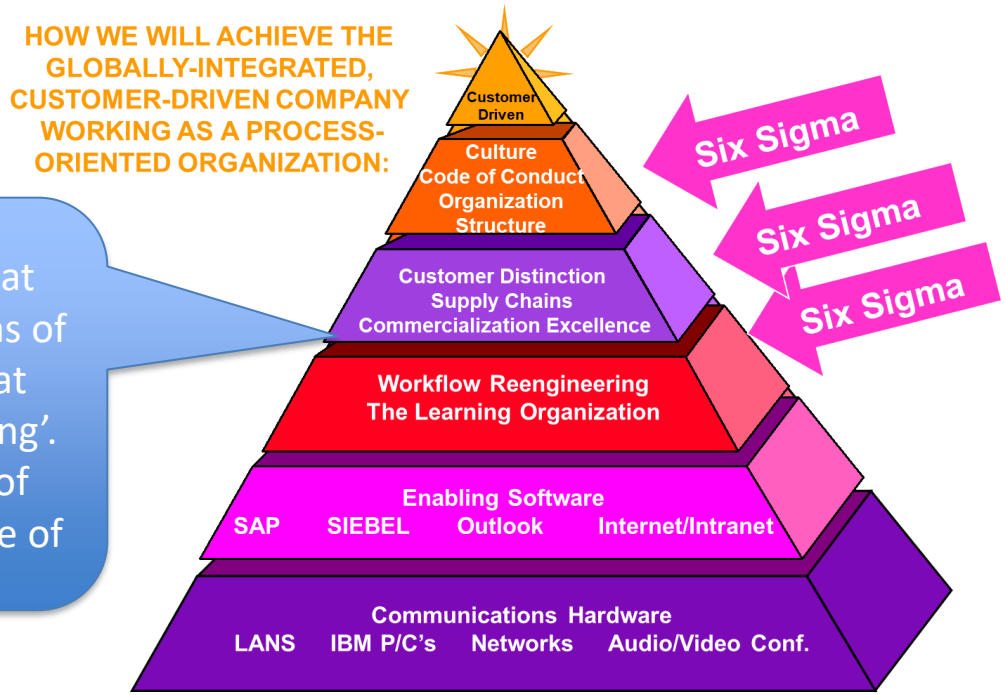
In the early 1990’s Dow Corning became embroiled in the Silicone Breast Implants saga with lawyers claiming that the implants caused systemic health problems. The claims first centered on breast cancer and then migrated to a range of autoimmune diseases, including lupus, rheumatoid arthritis and various neurological problems. This led to numerous lawsuits beginning in 1984 and culminating in a 1998 multibillion-dollar class action settlement. As a result, Dow Corning was in bankruptcy protection for nine years, ending in June 2004 during which time it largely withdrew from clinical markets.

Dow Corning's Transformation program

Prior to the SBI saga, Dow Corning had worked with IBM to develop their specific software requirement.
In 1994 they had started on a new Legal Entity Ordering Payments (LEOP). With the pressures of SBI and after spending 12 month and a significant amount of money in trying to develop LEOP unsuccessfully, Dow Corning decided to stop developing software and focus on what their strengths were and that was in Silicone products.
In 1995 the company announced that it would be implementing SAP. It also announced it program of transformation.

Dow Corning's approach of becoming a 'world class', customer focussed, and process oriented organisation

There was an acknowledgment that there were three areas of the organisation that needed 're-engineering'. Commercialisation of new products was one of them.



Commercialisation Issues

- Dow Corning was a heavy investor of Research and Design. It spent 10% of revenue on R&D.
- It's analytical capabilities was one of the best in the world. Both University of Michigan and Michigan State University would use its scanning electron microscope and other chemical analysis instruments.
- Prior to 1992, Research and Design was decentralised and there were R&D facilities at a number of locations around the world.
- After 1992, Dow Corning management created a central R&D location in Midland MI.
- Design and commercialisation of new products took on average 4 years.

Commercialisation Issues

- One of Dow Corning's Key Performance Indicators (KPI's) was New Product Market Share (NPS) - % of revenue that is from newly commercialised products.
- During the 1970's and 80's NPS was around 25%.
- By 1995 NPS had declined to 13%.
- Design and commercialisation of new products took on average 4 years.
- Dow Corning leadership determined that the way we commercialised products needed to improve.
- In 1997 it hired a consulting company to redesign its commercialisation process.

Commercialisation Issues

- Implementation of the new commercialisation process was not successful.
- Poor change management practices led to poor take up of the new processes.
- In 1999 Dow Corning's leadership decided to abandon the new commercialisation process and focus on the Lean Six Sigma implementation.
- In 2001 Dow Corning decided as part of its Lean Six Sigma program to implement Design for Six Sigma.

Business Leaders drive the three principal methods used in Implementing Six Sigma which are: The Improvement Process (DMAIC), Design for Six Sigma (DFSS), and Process Management

- DFSS generates new processes, products, services, and/or plants
- DMAIC improves existing process performance
- Process Management is the system that enables leverage and sustaining gains achieved by DFSS and SIGMA Improvement Process
- Leaders drive and align the efforts strategically



Six Sigma Design Process - DMEDI

	Objective	On-Going Activities	Main Activities	Potential Tools and Techniques	Key Deliverables
1.0 Define EXECUTE	• Understand the Purpose and the Outputs of Define • Develop a Charter for the Project • Develop a Multi-Generational Product/Service Plan • Develop a Project Plan to Manage the Project	Change Management	• Form Stakeholder Review Team • Develop Business Case • Develop Opportunity Statement • Develop Goal Statement • Develop Project Scope • Develop Multi-Generational Plan • Develop Project Plan • Develop Preliminary Project Charter • Identify Team and Support Resources • Develop Communication and Change Management Plan	Multi-Generation Plan; Charter; Gantt Chart; Communications Plan; In-and-Out-of-Scope Tool (In Scope; Out of scope; Out of scope)	
2.0 Measure EXECUTE	• Understand the purpose and outputs of Measure • Understand how to develop and specify CCRs using QFD and other methods • Begin QFD work for one project • Understand how to collect and use VOC & benchmarking information for design • Begin work on VOC plan for project • Synthesize information and create output from Measure	Customer Alignment; Strategy Alignment	• Research Customers • Identify Customer, Environmental / Health / Safety (EHS), and Business Needs • Translate needs into Critical to: Customer, EHS, and Business Requirements • Benchmark competitor performance • Organize information using QFD Matrix • Prioritize CCR's • Establish CCR upper and lower specifications • Establish Target Sigma Performance • Complete CCR Matrix • Project Review	Voice of the Customer; Affinity Diagram; Structure Tree; Benchmarking; Quality Functional Deployment; QFD; Critical To Customer Requirements Matrix	• Customers/Stakeholders & Segmentation • Voice of Customer • Competitive Benchmarking • Product/Service Requirements Definition • Critical-to-Customer Requirements (CCR) Definition • Business Risk Mgmt
3.0 Explore EXECUTE	• Understand the purpose and the outputs of Explore Step • Use a House of Quality to deploy CCRs to design "pieces" • Use creative techniques to develop alternative concepts • Evaluate conceptual designs • Develop high-level designs • Assess and select the best high-level design • Assess the risk of your high-level design	Project Management	• Translate CCR's into Functions • Identify Conceptual alternatives • Select best fit concept • Develop functional deployment map • Evaluate CCR's vs, Functions with QFD matrix • Perform Functional Capability Assessment • Translate Functional Requirements into High Level Design Elements • Evaluate Functional Requirements vs. High Level Design Elements using QFD matrix • Perform Sigma Capability Assessment • Project Review	Structure Tree; Function/Design Element QFD's; Simulation; Benchmarking; CCR/Function QFD; Capability Assessments; Creativity Tools; Functional Deployment Map; Flow Up; Pugh Matrix; Score Card	• Product/Service Design Concepts • "Best-Fit" Concept Selection • High-Level Production Process Design • Design Elements Definition • Functional Capability Assessment • Performance (Sigma) Capability Assessment • Score Card • Business Risk Mgmt
4.0 Develop EXECUTE	• Understand the purpose and outputs of Develop • Begin work on planning detailed design step activities • Develop a detailed product, service or process design • Refine capability predictions begun in the Explore step • Develop process control methods for the "production" process • Develop pre-pilot and pilot test plans for the new product/service	Project Case Development	• Translate High Level Design Elements into Detailed Design Elements • Develop Detailed Designs • Perform Capability Assessments • Revise design based on Capability Assessment • Perform FMEA & EMEA on design • Improve Design based on FMEA & EMEA • Develop Process Control Plans • Develop Pilot Plans • Project Review	Structure Tree; Detailed Designs; Simulation; Design Element QFD's; DOE; Capability Assessments; FMEA / EMEA; Weibull; Flow Up; Process Control Plans; Fault Tree; Score Card	• Detailed Product/Service Design • Detailed Production Process Design • Refined Functional Capability Assessment • Refined Performance (Sigma) Capability Assessment • Validation Plans • Process Control Plan • Score Card • Business Risk Mgmt
5.0 Implement EXECUTE	• Understand the purpose and the outputs of Implement • Develop and execute the pilot and analyze the results • Develop full-scale implementation plans and transition to process owners • Evaluate the design process and make improvements	Process Management; BUILD, ANALYZE, PLAN, FOCUS	• Execute the Pilot and Validation Plans • Analyze Results • Revise/Confirm Design • Scale Up Decision • Develop Detail Implementation Plans • Develop Communications Plans • Transition Design • Develop Process Owner Transition Plan • Develop Communications Plans • Complete Project Documentation • Document Lessons Learned • Rewards and Recognition • Project Review and Closure	Gantt Chart; Communications Plan; Pilot Validation Assessment; Score Card; Process Control Plans; Rewards & Recognition	• Validation Testing Complete • Gap Analysis/Redesign • Scale-Up Decision • Full-Scale Implementation Plan • Process Owner Transition Plan • Build & Control Documentation • Design Transitioned • Team Lessons Learned • Reward & Recognition • Business Risk Mgmt

Deck Protector Case Study

One of Dow Corning's major customers, ArmorAll©, approached us with the idea of taking on Thomspon's WaterSeal© with a Silicone Modified Organic (SMO) product.

As the number of permutations with the Siloxane backbone was limited, Dow Corning's strategy for growth was SMO. This would have been the first product that could have been commercialised.
ArmorAll© wanted us to develop and commercialise the product within 18 months!
This project became the pilot for the DfSS methodology.



Product Development

The Dow Corning chemist started to work on a product that would meet or exceed the competitors performance criteria, namely its water proofing.
The chemists came up with an SMO emulsion.
The next challenge was to find the right combination of Silane, Wax, and surfactant that would meet the price and performance requirements.
A six factor, two level, Design of Experiment was conducted. Samples were then tested for water proofing durability and results compiled with the relevant economic / profitability data.

Product Development

Process engineering work had commenced concurrently. A new process was designed based on volumes provided by ArmorAll©.
Once all the information was completed a key meeting was conducted with ArmorAll© to discuss the results and decide which formulation and price they would like to go with.
When the work was presented to ArmorAll© they were extremely impressed with the detail.
Agreement was made and work commenced to produce the estimated demand.

Results of the ArmorAll© Deck Protector

- In 12 months over 4,500 tonnes of product was produced and generated $US40 million of revenue.
- The project won a Technical Achievement Award and a Manufacturing Achievement Award.
- It became the basis for all new commercialisation project for Dow Corning.

Post Commercialisation

After 12 months of a very successful campaign the project stopped.
ArmorAll © had ran TV commercials that Thompson's challenged them on.
A legal battle ensued and ArmorAll © lost.
The financial impact of the court case forced ArmorAll © to be taken over by Chlorox .

Lessons Learned

- New Lean (Six Sigma) approach to commercialisation was a complete success.
- Opened up the use of complex statistical and mathematical modelling in the product development process.
- Existing processes were challenged and when identified as non valued added, were removed.
- No mater how good your commercialisation process, poor risk management by your customers can lead to project failure.

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Questions ?